determine its shape, you walk around a circle in the upper region measuring its circumference, and then walk through the center from one side of the circle to the other, measuring its diameter. If your universe were flat, then the circumference would be π = 3.14159... times the diameter. But the circumference, you discover, is far smaller than the diameter. Your universe, you conclude, is highly warped!

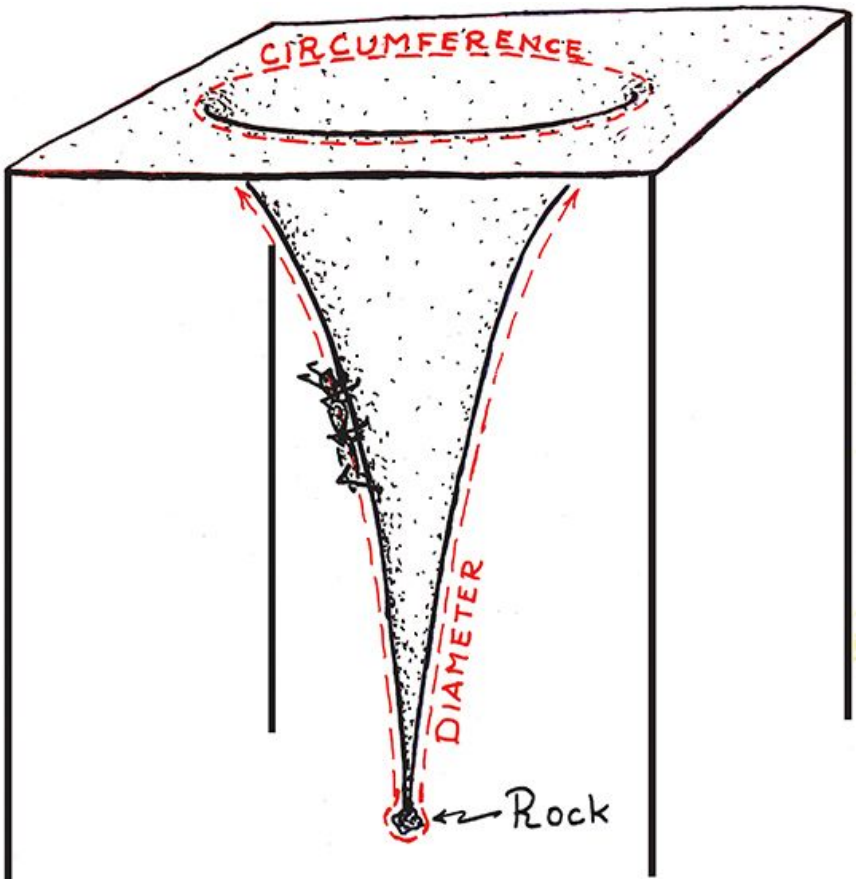


Fig. 5.1. An ant on a warped trampoline. *[My own hand sketch.]*

Space around a nonspinning black hole has the same warping as the trampoline: Take an equatorial slice through the black hole. This is a two-dimensional surface. As seen from the bulk, this surface is warped in the same manner as the trampoline. Figure 5.2 is the same as Figure 5.1, with the ant and poles removed and the rock replaced by a *singularity* at the black hole's center.